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MILITARY SPECIFICATION

SERVICING UNITS, COMPRESSED GAS, MOBILE

This specification is approved for use within the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of compressed gas servicing units (hereinafter called "servicing units") for mounting on either a M116, 3/4 ton trailer chassis or on a suitable commercial chassis as specified; one system for air and one for dual air/oxygen servicing. Each servicing unit will carry eight high-pressure supply cylinders and is to be used for recharging one to four SCUBA cylinders at the same time.

1.2 Classification. The servicing units shall be of the following types as specified (see 6.2):

Type I - Air
Type II - Dual air/oxygen

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATONS

FEDERAL

RR-C-901	- Cylinders, Compressed Gas: With Valve or Plug and Cap; DOT 3AA.
UU-T-81	- Tags, Shipping and Stock.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-R-12144	- Reflector, Indicating Clearance.
MIL-T-22085	- Tape, Pressure-Sensitive, Adhesive, Preservation and Sealing.
MIL-C-45150	- Chassis, Trailer, 2-Wheel, Cart Type, 1/4- to 3-1/2-Ton.
MIL-C-52437	- Chassis, Trailer, Single-and-Multi-Axle.
MIL-C-83960	- Cylinder, Oxygen Purifier.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standard for Federal Service.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria For Military Systems, Equipment and Facilities.
MS51331	- Guard, Splash, Wheel.
MS53028	- Chassis Trailer, 3/4 Ton, 2 Wheel, Military Design, M116A1.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations

49 CFR parts 171-179 - Hazardous Materials Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DRAWINGS

ME

13227E5951 - Valve, Ball, Quarter Turn.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

ANSI/AWS D1.1 - Structural Welding Code, Steel.
ANSI/AWS D1.2 - Structural Welding Code, Aluminum.

(Application for copies should be addressed to the American Welding Society, Inc., 550 North LeJeune Road, Miami, FL 33126.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The servicing unit shall consist of a body assembly for mounting on a Government-furnished M116, 3/4 ton, trailer chassis (see 3.14) or on a commercial trailer (see 3.9.1.4). The body shall support eight standard DOT 3AA cylinders on a rack in a horizontal position so that the nominal 51 to 55 inch by 9 inch diameter cylinders can be slid in or out from the rear and secured in place against a bulkhead. A gas flow control system shall be located forward of the bulkhead and shall consist of piping, manifolds, fittings, valves, gages, safety devices, purifier, and a control panel that shall provide a complete servicing system when connected to SCUBA cylinders (see figures 1 and 4 for flow diagrams). A gas flow control system panel shall be provided. The type I servicing unit shall be for air use only, and shall contain the components shown in figures 1 and 3. The type II servicing unit shall be for dual air and oxygen service and divided as shown in figures 4 and 5.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of servicing units from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish servicing units conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The item(s) shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide

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protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the servicing units may be newly fabricated from recovered materials to the maximim extent practicable, provided the servicing units produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the servicing units.

3.4 Environmental conditions.

3.4.1 Operation temperature. The servicing units shall perform as specified herein in any ambient temperature from +120° F to -25° F.

3.4.2 Storage temperature. The servicing units shall not be damaged by storage in any ambient temperature from +155° F for 4 hours daily to -65° F without solar radiation for 12 hours.

3.5 Safety. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. The servicing unit shall be constructed such that it will comply with applicable sections of the hazardous materials regulations of the Department of Transportation (49 CFR parts 177.834 and 177.84).

3.6 Human factors engineering design requirements. The servicing unit shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, 4 (General Requirements), 5.1 (Control/Display Integration), 5.2 (Visual Displays), 5.4 (Controls), 5.5 (Labeling), 5.6 (Anthropometry), 5.9 (Design for Maintainability), and 5.13 (Hazards and Safety), of MIL-STD-1472, as applicable.

3.7 Maintainability.

3.7.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.5) shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.7, and shall not require scheduled maintenance more often than every 100 hours of operation.

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3.7.2 Maintenance ratio. The servicing units shall have a maintenance ratio of not more than 0.04, when tested as specified in 4.5.2.7.2. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.8 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners.

3.9 Construction.

3.9.1 Design. The servicing unit shall be designed to mate with either the M116, 3/4 ton, trailer chassis conforming to MIL-C-45150 and MS53028 or when (see 6.2), with a commercial trailer chassis conforming to MIL-C-52437, size 4 (see 3.9.1.4). The weight of the servicing unit shall be evenly distributed to the trailer chassis frame. The servicing unit shall be structured so that there is no need for torsion, moment, or other stress support from the trailer chassis; and, so that no damaging, deforming, or otherwise excessive torsion, moment, or other stresses are imparted to members of the trailer chassis from the servicing unit during operation and towing tests specified herein. Unless otherwise specified herein, no modification of the Government-furnished trailer is permitted. Not less than six mounting points shall be utilized to secure the servicing unit to the chassis.

3.9.1.1 Weight. The total weight of the servicing unit, cylinders and trailer chassis shall not exceed 3300 pounds with the M116 trailer chassis and 4000 pounds with the commercial trailer chassis.

3.9.1.2 Center of gravity. When mounted on the trailer chassis and loaded with a full complement of cylinders, the horizontal center of gravity of the servicing unit shall be such that a vertical force of not less than 100 pounds nor more than 200 pounds shall be imposed on a towing pintle by the trailer lunette with the trailer level.

3.9.1.3 Compatibility. The servicing unit, mounted on the M116 trailer chassis, or on a commercial trailer chassis (see 3.9.1.4), shall be compatible for towing behind any 3/4 through 2-1/2 ton military design truck, including the M37 and the M35 cargo trucks as well as other self-propelled military equipment.

3.9.1.4 Commercial trailer. When specified (see 6.2), a commercial trailer conforming to MIL-C-52437, size 4, option A (flotation) shall be provided. A 24-volt military electrical system shall be provided (T-14 in MIL-C-52437). All provisions of MIL-C-52437 except the bed size, and testing which shall be in accordance with those provisions selected in section 4 shall be followed. The bed size and mounting holes shall be as shown in figure 7 and MS53028.

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3.9.2 Cylinder rack. The cylinder rack shall accommodate eight cylinders lengthwise on the trailer-mounted unit as shown in figure 2. There shall be individual troughs and adequate supporting framework with provisions to prevent the accumulation of water around the cylinders when the trailer is in its normal position. Each cylinder shall be removable separately without moving the others and provisions shall be made to permit loading of the cylinders with rollers or other devices. The rack, with bulkhead and clamping, shall securely hold the compressed gas cylinders against movement during the operation and mobility tests specified herein.

3.9.2.1 Clamps. A quick-acting and easily accessible clamping arrangement, designed to hold the cylinders in place, shall be provided integral with the cylinder rack. Clamps shall not require tools to operate.

3.9.2.2 Bulkhead. A cylinder bulkhead with openings for cylinder necks shall be provided at the front of the cylinder rack.

3.9.3 Gas flow control system components. The component piping, fittings, valves, gages, purifier(s), regulator(s), and connecting lines of the gas flow control system shall be designed for a working pressure of not less than 3500 pounds per square inch gage (psig). They shall be constructed of materials entirely suitable for the purpose, cleaned and shall be compatible for the gas service intended. Safety devices shall be provided at each manifold, at each regulator, and as required to properly protect the compressed gas system. The gas flow control system shall be capable of withstanding a hydrostatic pressure of not less than five-thirds of the working pressure. All components of the gas flow control system shall be delivered in an oxygen clean state.

3.9.3.1 Fittings. When specified herein; gas fittings on each of the two types of gas flow control systems shall be in accordance with FED-STD-H28, as follows: Connection No. 540 for oxygen service and connection No. 1340 for air services.

3.9.3.2 Manifold system. The cylinders shall be manifolded as shown in figure 1 for the type I servicing unit and as shown in figure 4 for the type II servicing unit. Each cylinder shall be connected to a pressure gage and a cam-lever-operated valve with a flexible metal-wrapped hose.

3.9.3.2.1 Flexible connecting lines. Flexible metal-wrapped hoses for high-pressure gas service shall be furnished as connecting lines from the storage cylinders to the manifold. Fittings at each end shall be as specified in 3.9.3.1.

3.9.3.2.2 Individual pressure gages. Individual moistureproof pressure gages shall be provided for each cylinder. The gages shall be suitable for high-pressure gas service. The gages shall have a pressure range of 0 to not less than 3500 pounds per square inch (psi) and shall be not less than 2-1/2 inches in size. The gages shall be a part of the manifold system and shall be mounted between the cylinders and cam-lever-operated valves (see figure 1). Cylinder-side fittings shall be as specified in 3.9.3.1 to mate with the flexible lead

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fittings. Gages for oxygen service shall be clearly labeled. The gages shall not be interchangeable with the gages on the air side of type II servicing unit panels.

3.9.3.2.3 Quarter turn ball valves. Valves shown in figures 1 and 4 as quarter turn ball valves shall be 2-way, straight through design quarter turn ball valves. The valves shall conform to the dimensional limits shown in 13227E5951. One valve shall be provided for each cylinder. Each valve shall provide positive shutoff to working pressure (3500 psi) air or oxygen from the manifold side and the cylinder side of the valve. The piping connections shall be 1/4 inch female NPT, in the arrangement shown in 13227E5951. Seats in the valves shall be polychlorotrifluoroethylene. The valve body shall be 316 stainless steel. Handles shall be sized and spaced in accordance with MIL-STD-1472, rotary controls. The on and off positions shall be spaced 90 degrees apart and they shall be clearly marked. Preferably the valve handle will point up to indicate on position.

3.9.3.2.4 Recharging valve. A recharging valve shall be incorporated in the manifold system to provide facility for recharging the storage cylinders without removing them from the servicing unit. The valve shall have design working pressure of 3500 psig and an outlet connection as specified in 3.9.3.1.

3.9.3.3 Purifier. A cartridge-holding-type MB-1 cylinder conforming to MIL-C-83960 shall be incorporated as shown in figure 1 for type I servicing units, and as shown in figure 4 for type II servicing units.

3.9.3.4 Manifold shutoff valve. A manifold shutoff valve shall be located between the manifold and the purifier cylinder as shown in figure 1 for type I servicing units and as shown in figure 4 for type II servicing units.

3.9.3.5 Regulators. Regulators shall be provided. They shall be designed for a normal working pressure of 3500 psig and shall be adjustable to any outlet pressure setting between 50 psig and a maximum of not less than 3000 psig, inclusive. Each regulator shall incorporate features which protect it and its pressure control capability against inlet pressure surges. The regulator shall allow gas flow in only one direction so as to prevent backflow when pressure in the SCUBA charging manifold is greater than that in the supply cylinder manifold. The regulator shall not emit gas to the atmosphere by such features as back-venting or self-venting.

3.9.3.6 Service pressure gage. A service pressure gage shall be provided to indicate the servicing and regulator outlet pressure. The gage shall have a scale range of 50 to not less than 3000 psi. The dial size shall be no less than 4 inches. One gage as shown in figure 1 for type I servicing units and two gages as shown in figure 4 for type II servicing units.

3.9.3.7 Servicing valve. Servicing valve(s) shall be located between the regulator and the SCUBA charging manifold: One valve as shown in figure 1 for type I servicing units, and two valves as shown in figure 4 for type II servicing units.

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3.9.3.8 SCUBA charging manifold. A manifold as shown in figure 3 shall be provided for simultaneous charging of four SCUBA cylinders for type I servicing units. Two manifolds as shown in figure 4 shall be provided for simultaneous charging of four SCUBA cylinders (two with air, two with oxygen) for type II servicing units. For type I servicing units, each manifold shall have a bleed-off valve and a 0 to 3000 psi, inclusive, scale pressure gage. Flexible metal-wrapped hose charging leads and yoke-type filling adaptors shall be provided for connecting to standard "J" and "K" type SCUBA cylinder valves. The yoke adaptorsshall be of the type having integral bleed-off valves. For type II servicing units, the oxygen manifold shall have a bleed-off valve and a 0 to 3000 psi, inclusive, scale pressure gage. Flexible metal-wrapped hose charging leads and yoke-type filling adaptors shall be provided for connecting to standard medical port outlet connection No. 870. The yoke adaptors shall be of the type having integral bleed-off valves.

3.9.3.8.1 SCUBA cylinder trough. A trough shall be provided to hold water for cooling of SCUBA cylinders during charging. The trough shall be constructed of corrosion resistant materials or materials which are treated for corrosion resistance. A drain outlet with plug shall be included. The trough shall be 31 inches in depth and shall be a minimum of 12 inches wide by 40 inches long or may be constructed in two sections. Provisions shall be made for mounting the trough on the trailer.

3.9.3.9 Control panel. A control panel shall be mounted forward of the storage cylinders at a distance sufficient to provide accessibility for operation and maintenance. The valves, gages, manifold, regulator(s), purifier(s), and connecting lines shall be securely mounted on the panel. All components mounted on the panel shall be positioned so as not to interfere with or obstruct the use or removal of adjacent items. No component or portion thereof shall extend beyond the top or outer edges of the panel. The components of the dual gas system (type II) shall be mounted on separate halves of the control panel as shown in figure 5.

3.9.4 Shrouding. All components forward of the cylinder rack bulkhead shall be protected by an all-enclosing shroud. The shroud shall be a sheet metal cabinet with doors as needed for access to the control panel and to the cylinder valves. No tools shall be needed for access to rear of panel or to cylinder valves.

3.9.4.1 Undershield. An undershield shall be provided if a functionally similar assembly is not otherwise included. It shall cover the complete floor area covered by the shroud and shall protect the control panel and gas flow control system against stones and other foreign objects.

3.9.5 Toolbox. A hinged-top metal box for storing repair parts and maintenance and operating tools shall be furnished on the servicing unit body, and shall be easily accessible.

3.9.6 Fenders. The servicing unit shall be equipped with fenders and splash guards conforming to SAE J321, J682 and J683 which shall restrict water and

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other road-surface substances thrown from the trailer wheels. The fenders shall be contour-shaped to the trailer's normal axle centers or of octagonal shape about the trailer wheels, and shall include splash shields. Each fender shall support a minimum of 200 pounds resting on any one square foot area without permanent deformation. The front, rear, and outer edges of the fenders shall be rolled or angled to prevent brush damage and reduce road splash. The splash guards shall be dimensionally interchangeable with MS51331 and incorporate antisail characteristics. The splash guards shall be of flexible fiberglass reinforced plastic, polyester or solid polyvinyl types which shall withstandcold weather to -25° F. The minimum physical properties of the splash guards shall provide a tear strength of not less than 375 lbs/in and an ultimate strength of not less than 3000 psi.

3.9.7 Rear stabilizing jack. The servicing unit shall be equipped with a gear-type, single-leg jack assembly that shall function as a safety stabilizing stand. The jack shall be fastened to the center of the rear crossmember of the trailer in accordance with the jack manufacturers' recommendations, and shall not affect the removal or the installation of the compressed gas supply cylinders when in the parking-stand position. The jack shall have a positive lock in both the parking-stand position and the travel position. The jack shall be equipped with a handcrank, gearbox, telescopic leg, and mud-pad foot. The handcrank shall have a throw of not less than 5-1/2 inches and the gearing ratio shall provide not less than two crank-handle revolutions for each inch of leg travel. The mechansim shall be protected from foreign matter which would impair functional efficiency. In the parking-stand position, the jack shall have a travel distance of not less than 18 inches from fully retracted to fully extended. When retracted, the mud-pad shall be no more than 18 inches below the bottom of the trailer's rear crossmember and shall be no less than 32 inches below the rear crossmember when fully extended. In the travel position, the jack assembly shall not reduce the departure angle to less than 30 degrees.

3.9.8 Reflectors. Reflectors conforming to MIL-R-12144 shall be furnished and mounted as required by DOT Motor Carrier Safety Regulations. In addition, two amber reflectors shall be provided on the front of the control panel shroud at the same height as those mounted on the sides.

3.9.9 Mobility. The trailer, mounted with the servicing unit and a full complement of cylinders, shall perform as specified herein with no evidence of damage, permanent deformation displacement of any component, accessory, or part installed in or on the body or chassis. There shall be no evidence of defective welded, bolted, or riveted connections.

3.9.9.1 Trailing and turning. The trailer-mounted servicing unit shall follow the towing truck without bottoming, weaving or sidesway. Turns to the right and to the left up to an angle of 60 degrees from straight ahead shall not cause interference between or damage to the truck of the trailer-mounted servicing unit.

3.9.9.2 Braking. The trailer-mounted servicing unit shall withstand sudden stops from an initial speed of 50 mph over primary highways and from speeds of up to 10 mph over cross-country terrain.

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3.10 Slinging and tiedown provisions.

3.10.1 Slinging provisions. Permanently affixed slinging provisions that enable the servicing unit to be lifted in its normal travel position shall be provided and shall conform to MIL-STD-209. Slinging provisions may also be used as tiedowns when such provisions conform to MIL-STD-209.

3.10.2 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the servicing unit to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209. Tiedown provisions may also be used as slinging provisions when such provisions conform to MIL-STD-209.

3.11 Identification marking and shipping data.

3.11.1 Identification marking. The servicing unit shall be identified in accordance with MIL-STD-130. The marking shall be appplied to the servicing unit on plates conforming to MIL-P-514, type I, style 3, class 2, composition A, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plate shall be both visible and legible.

3.11.2 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III, class 2, composition A, of type I, grade A, class 1 material shall be provided. In addition, the plate shall show the silhouette of the servicing unit in transport position indicating the center of gravity and the location and capacity of the slinging and tiedown provisions. The plates shall be attached by screws, bolts or rivets in a location where the plates shall be both visible and legible.

3.12 Instruction plates. Each servicing unit shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating or servicing the servicing unit. Instruction plates shall conform to MIL-P-514, type III, class 2, composition A, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts or rivets in a location where the plate shall be visible and legible.

3.12.1 Warning plates. Warning plates (see 3.13) shall be prominently displayed or both front and rear sides of the panel of the type II servicing units (see figure 6).

3.13 Treatment and painting. The portions of the servicing unit normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, type F or G Green 383 (see 6.2), except for the control panel of the type II service unit. The air servicing half panel and the oxygen servicing half panel shall be painted with finish colors conforming to MIL-T-704: Black, No. 17038 for air, Green, No. 14187 for oxygen.

3.14 Government-furnished property. When specified (see 6.2), the following property in the quantities indicated will be furnished by the Government (see 6.4):

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Item No.	Description	Identification	Quantity for each servicing unit	
			Type I	Type II
1.	Trailer, 3/4 ton M116	MIL-C-45150 MS53028	1	1
2.	Cylinder, Compressed gas, air, ICC 3AA, 3500 psi	NSN 8120-00-842-3566 or NSN 8120-00-821-2645 or NSN 8120-00-806-4579 RR-C-901	8	4
3.	Cylinder, Compressed gas, oxygen, DOT 3AA, 2265 psi.	NSN 8120-00-285-1576 RR-C-901	0	4

3.15 Transportability.

3.15.1 Rail. The unpackaged servicing unit shall withstand rail humping without damage or permanent deformation to any part of component when tested as specified in 4.5.2.8. After testing, the servicing unit shall operate as specified herein for a period of not less than 1 hour without malfunction or permanent deformation of any part of the servicing unit.

3.16 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characterisitcs as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance. When specified (see 6.2), interchangeability between type I and type II panels and parts for the same gases shall be required.

3.17 Noninterchangeability. On type II servicing units, like parts (gages, lines and fittings) shall not be interchangeable between the air and oxygen systems.

3.18 Workmanship.

3.18.1 Castings and forgings. All parts, components and assemblies of the servicing unit which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be liminted to procedures which do not reduce mechanical properties or affect function.

3.18.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel may be employed

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instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.18.3 Welding.

3.18.3.1 Aluminum welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded arc method. Welding procedures shall be qualified in accordance with ANSI/AWS D1.2 or ASME as given in 3.18.4. Welding shall be in accordance with one of these two documents.

3.18.3.2 Steel welding. The surfaces or parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the servicing unit to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the servicing unit shall be at the option of the contractor. Welding procedures shall be qualified in accordance with ANSI/AWS D1.1 or ASME as specified in 3.18.4. Welding shall be in accordance with one of these two documents.

3.18.3.3 General Welding Practice. Welds may be ground, filed, buffed or chipped but shall not be hammered or peened, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, overlap, abrupt weld face ridges, lack of prescribed fit or spatter. A weld may be repaired once by rewelding. Any additional rewelding or weld metal deposition shall require prior approval by the government quality assurance representative. All joint preparation and welding for repair shall comply with the original requirements for the joint. Welding procedures qualification and repair procedures shall be made available for review by the contracting officer or contracting officer's representative.

3.18.4 Welders and welding operators. Before assigning any welder to manual welding work or operator to automatic welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/AWS D1.1, Structural Welding Code, Steel.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.2, Structural Welding Code, Aluminum.

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Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The certification shall be made available for review by the contracting officer or the contracting officer's representative.

3.18.5 Welding practices.

a. Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the supplier's recommendations.
b. Tack welds shall be subject to the same quality requirements as final welds.
c. Work shall be positioned for flat welding whenever practicable.
d. Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part and the servicing unit.
e. Repair welds shall be subject to the same quality as the specifications call for.

3.18.6 Weld quality.

a. The contractor shall be responsible for determining the inspection needed to insure that, when the welded parts are assembled together to make the servicing unit, the servicing unit shall conform to the inspection requirements specified in section 4, as a minimum, 100 percent visual inspection of the welds.
b. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.
c. Undercut in weldments shall be held to a minimum and shall be not more than 1/32 inch deep, or 10 percent of the parent metal thickness, whichever is less.

3.18.7 Brazing. Brazing of steels, copper alloys, and nickel alloys shall conform to MIL-B-7883.

3.18.8 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

3.18.9 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. When a commercial trailer is required, the first article shall also be subjected to the examinations specified in MIL-C-52437. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first article servicing unit shall be tested as specified in table I. All tests shall be performed with the manifold mounted on the trailer.

TABLE I. Tests.

Inspection	Requirement	Test Method
Construction.	3.9	4.5.2.1
Design.	3.9.1	4.5.2.2 and subparts 2/
Gas flow control system components.	3.9.3	4.5.2.3
Hydrostatic pressure	3.5	4.5.2.4

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TABLE I. Tests. (Cont'd)

Inspection	Requirement	Test Method
Tiedown provisions.	3.10	4.5.2.5 and subparts
Slinging provisions.	3.10	4.5.2.6 and subparts
Maintenance support.	3.7.1	4.5.2.7 and subparts
Rail.	3.15.1	4.5.2.8 and subparts
Operation conditions.	3.4.1	4.5.2.9 1/
Lubrication.	3.18	MIL-C-52437, 4.6.2.1 1/
Stress Analysis.	3.4.1	MIL-C-52437, 4.6.2.2 1/
Road runs.	3.7.2	MIL-C-52437, 4.6.2.3 1/
Brakes.	3.7.2	MIL-C-52437, 4.6.2.4 1/
Fording.	3.7.3	MIL-C-52437, 4.6.2.5 1/
Failure criteria.	3.4.1	MIL-C-52437, 4.6.2.6

1/ Test to be performed when commercial trailer is specified only (see 6.2 j).
2/ Test to be performed only when commercial trailer not specified.

4.4 Quality conformance inspection.

4.4.1 Examination. Each servicing unit shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each servicing unit shall be subjected to the tests specified in 4.5.2.1, 4.5.2.2, 4.5.2.2.1 except mileage shall be decreased to 10 miles for each course, 4.5.2.3 and 4.5.2.4. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The servicing units shall be examined as specified herein for the following characteristics:

Critical

1. Safety devices not as specified.
2. Leaks in pressure lines.
3. Any piping not oxygen clean as specified.

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.

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103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the servicing units.
106. Dimensions and weight not as specified.
107. Parts or components missing.
108. Assembly incomplete or incorrect.
109. Fittings and fasteners not as specified.
110. Human factors not as specified.
111. Ease of maintenance not as specified.
112. Welder, welder operator, and weld procedure qualifications not as specified.
113. Welding not as specified.
114. Castings, forgings, and metal fabrication not as specified.
115. Brazing not as specified.
116. Bolted and riveted connections not as specified.
117. Safety provisions not as specified.

Minor

201. Treatment and painting not as specified.
202. Identification and other required markings not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. The operation and hydrostatic tests shall be conducted after completion of the mobility tests.

4.5.2.2 Mobility. Perform all mobility tests with a fully loaded servicing unit (8 full cylinders, capped, and any other accessory equipment in place on the trailer.) At the completion of each mobility test, examine the servicing unit and the trailer chassis. Evidence of damage to or displacement of any component, part or accessory, or evidence of defective welded, bolted or riveted connections shall constitute failure of these tests.

4.5.2.2.1 Road. Tow the trailer-mounted unit over roads as specified below. Subject the trailer-mounted unit to at least 5 stops as specified in 3.9.9.2 during each phase of the test, and trail and turn as specified in 3.9.9.1. Nonconformance to 3.9.9 shall constitute failure of this test.

Course	Mileage and Speeds
Hard surface (highway)	25 miles at variable speeds up to 50 mph.
Cross-country (rough terrain not previously leveled or traveled)	50 miles at variable speeds up to 10 mph.

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4.5.2.2.2 Chatter course. Tow the trailer-mounted servicing unit over a chatter course until at least 2600 (either upward or downward) accelerations of magnitudes between 4G and 7G have been applied to a wheel center. The course shall be constructed so that the acceleration magnitudes can be obtained by adjusting the vehicle speed without the frequency of the accelerations exceeding 15 cycles per second (cps) or becoming less than 3 cps. The maximum acceleration magnitude shall not be exceeded, and accelerations having magnitudes less than the minimum shall be disregarded. The course shall be of such length that a wheel center receives, in one pass over the course, at least 50 consecutive vertical acceleration cycles of any magnitude less than the maximum specified herein. The accelerations shall be measured by a trial test to establish the required speed. Disconnect the recording instruments and complete the test at established speed. The acceleration valves used shall be those of the wheel center receiving the highest accelerations. Means for assuring that the test speed is held constant throughout the test shall be provided. Accelerations shall be measured on a running-gear member (usually an axle or spindle) that has a motion identical to that of the wheel center.

4.5.2.2.3 Diagonal V-ditch course. Tow the trailer-mounted servicing unit over a V-ditch course until 600 reversal torsional frame stresses have been applied. The B-ditches shall be a minimum of 18 inches deep, 36 inches wide at the top, at 45 degrees to the centerline of the course, and spaced a sufficient distance apart to allow only one wheel in the ditch at any time.

4.5.2.3 Operation. Operate and examine the servicing unit to insure the proper operation of the manifold system, valves and all other components. The unit shall be completely set up for operation using the rear stabilizing jack, the SCUBA cylinder trough, and other accessory equipment. Charge at least eightSCUBA cylinders, operating all components of the gas flow control system, including cascade valving. Allow the servicing unit to remain pressurized at 2400 psig for 1 hour and check for leaks (soap solution) over the entire piping and gas flow component system. Failure of any component or accessory equipment to function or any evidence of leaks or inability to charge SCUBA cylinders shall constitute failure of this test.

4.5.2.3.1 Valve back pressure check. With all cylinders in an empty state (0 psi) and the manifold recharging valves, the manifold shutoff valves and the quarter turn ball valves in the closed position, the maniflod filled with distilled water, attach an air source of 5800 psi (5/3 working pressure) to the recharging fitting. Open the recharging valve fully and leave the manifold in this state for 15 minutes. Any increase in pressure in the cylinder, as evidenced by the gauge readings or any leakage through the quarter turn valves, shutoff valves, or any fittings shall constitute failure of this test. All manifolds, 1 on type I and 2 on type II, shall require this test. Note: Type II oxygen fitting is different from the air manifold recharging fitting, see 3.9.1. Air may be used as a test gas in the oxygen manifold if the manifold and valves are returned to an oxygen clean state.

4.5.2.4 Hydrostatic. Subject the servicing unit, less cylinders, to a hydrostatic pressure of not less than 5800 psig (5/3 of 3500). The cylinder leads

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shall be plugged leak-tight. Hold the test pressure for 2 minutes. Any distortion, damage, or evidence of any leakage shall constitute failure of this test.

4.5.2.5 Tiedown provisions.

4.5.2.5.1 Test procedure. With the servicing unit anchored by means other than the tiedown being tested, apply the static leads specified in MIL-STD-209 to each provision. Measure the dimensions of the tiedowns.

4.5.2.5.2 Failure criteria. Weld failure, permanent deformation of the servicing unit or its tiedown provisions, breakage of the provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.6 Slinging provisions.

4.5.2.6.1 Test procedure.

a. Lift the servicing unit and hold it in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the servicing unit. Determine the forward angles of application for each provision. Measure the eye openings and clearance dimensions of the provisions.
b. Restrain the servicing unit by anchoring the main frame and subject each slinging provision to a force of 2-1/2-times the load it carries when suspended. This force shall be applied in the direction as determined in "a." above. An acceptable alternate method of test shall be to lift the servicing unit as in "a." above and add weights to the servicing unit until the force on each provision is 2-1/2-times the initial force. Hold the servicing unit under either of the above loading conditions for not less than 90 seconds.

4.5.2.6.2 Failure criteria. Weld failure, permanent deformation of the servicing unit or any slinging provisions, slings converging more then 24 feet above the lowest extremity of the servicing unit, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.5.2.7 Maintainability.

4.5.2.7.1 Maintenance support evaluation. Maintenance operations required during initial production testing shall be accomplished to determine conformance to 3.7.1. Failure of the support elements to maintain the servicing unit as specified shall constitute failure of this test. Errors or inadequancies in the manuals shall not be considered in assessing the maintenance ratio.

4.5.2.7.2 Maintenance ratio. The maintenance ratio shall be computed during all first article testing. Nonconformance to 3.7.2 shall constitute failure of this test.

4.5.2.8 Railroad car impact.

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4.5.2.8.1 Test procedure. Anchor the unpacked servicing unit on a domestic service railway freight car (blocking is permitted). Hump the car, with the servicing unit tied down, at speeds of not less than 4, 6, and 8 mph, respectively, against 5 empty freight cars with their brakes set and draft gear extended. The servicing unit shall face in the direction of the 5 cars. Repeat the procedure at not less than 8 mph with the rear of the servicing unit in the direction of the 5 cars. The servicing unit may be reversed by either reversing the direction of the flatcar (preferred) or by reloading the servicing unit in the proper direction on the freight car. Once the test has begun, there shall be no readjustment of the load, nor any reconditioning of the blocks, chocking material, or tiedowns, except as necessary to reverse the direction of the servicing unit. After each impact, the servicing unit shall be inspected. After completion of the railroad impact test, an operational checkout of the servicing unit shall be conducted. Nonconformance to 3.15.1 shall constitute failure of this test.

4.5.2.9 Environmental.

4.5.2.9.1 High temperature. The servicing unit shall be stored in an ambient temperature of +155° F for 4 hours. Care should be taken if compressed gas cylinders are equipped with fusible plug safety devices. Cylinders may be removed at the supplier's discretion while the temperature is above 140° F, but shall be installed for all other test temperatures. The servicing unit shall then be stabilized at 120° F. While at this temperature and in an equal ambient temperature, the servicing unit shall be operated as in 4.5.2.3. Malfunction of gages and controlling devices, inaccessibility of controls, damage to any components, or performance not in accordance with 3.4.1 shall constitute failure of this test.

4.5.2.9.2 Low temperature. The servicing unit shall be stored in an ambient temperatue of -65° F for not less than 12 hours and examined for damage. the servicing unit shall then be stabilized at -25° F and operated as in 4.5.2.3. During the test, all gages and controls shall be checked for proper operation. The accessibility of controls to operators wearing heavy clothes and mittens shall be checked. Malfunctioning of the gages and controls, inacessibility of controls, damage to any components, or performance not in accordance with 3.4.1 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inpsection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Examination. Each pack shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

118.	Openings not sealed as specified for level A or commercial.
119.	Protective metal cover not positioned and secured as specified for level A or commercial.
120.	Tags not attached as specified for level A.

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121. Marking on tags not applied with waterproof ink as specified for level A.
122. Technical publications not preserved as specified for level A.
123. Marking missing, illegible, incorrect, or incomplete for level A or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. All external openings into the trailer mounted servicing units shall be sealed with barrier material conforming to MIL-B-121, type I, grade A, class 1. The barrier material shall be secured in place with tape conforming to MIL-T-22085, type II. The protective metal cover shall be positioned to cover the control panel, then secured in a manner to prevent accidental opening. The servicing unit shall be tagged to indicate that all openings into the unit are sealed and that the seals must be removed prior to putting the servicing unit into service. Tags shall conform to UU-T-81, type A. Printing on tags shall be with waterproof ink. The trailer chassis components shall be preserved and packaged in accordance with MIL-STD-281, level A, for mobile shipment. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3 and placed in the toolbox.

5.1.2 Commercial. Each complete trailer mounted servicing unit, with the external openings sealed and the cover over the control panel secured, shall be preserved in accordance with ASTM D 3951.

5.2 Packing (see 6.7). Each complete trailer mounted servicing unit, preserved as specified in 5.1, shall be prepared for mobile shipment in accordance with carrier rules and regulations.

5.3 Marking.

5.3.1 Military. Marking for level A shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition to the marking required, the cube and weight shall also be indicated.

6. NOTES

6.1 Intended use. The servicing unit is intended for refilling SCUBA cylinders in remote field sites.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type required (see 1.2).
c. Time frame required for submission of first article model (see 3.2).

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d. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and test will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
e. Color required (see 3.13).
f. When the Government-furnished property is required (see 3.14).
g. When interchangeability between type I and type II panels is required (see 3.16).
h. When welding process other than as specified is required (see 3.18.3.2).
i. Degree of preservation required (see 5.1).
j. When a commercial trailer chassis shall be provided (see 3.9.1).

6.3 Preproduction model. Any changes or deviations of production servicing units from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish servicing units conforming to this specification.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.14, when applicable (see 6.2).

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of servicing units.

6.6 Packing. Degrees of packing have not been included as the packing specified is applicable for all degrees of packing. For purposes of preservation/packing level marking the packing shall indicate level A.

Custodian:
Army - ME

User activity:
Army - ER

Preparing activity:
Army - ME

Project 3655-A100

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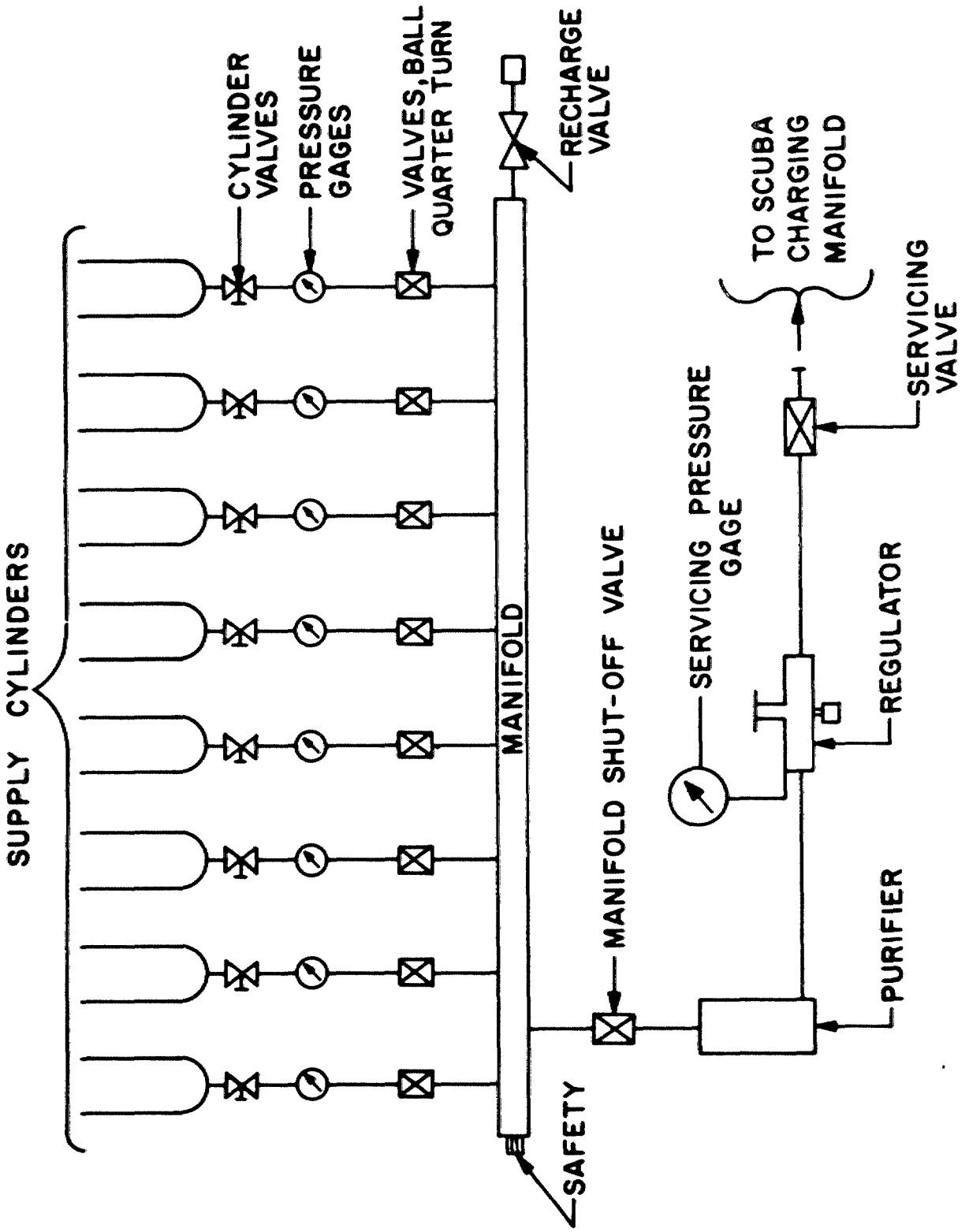
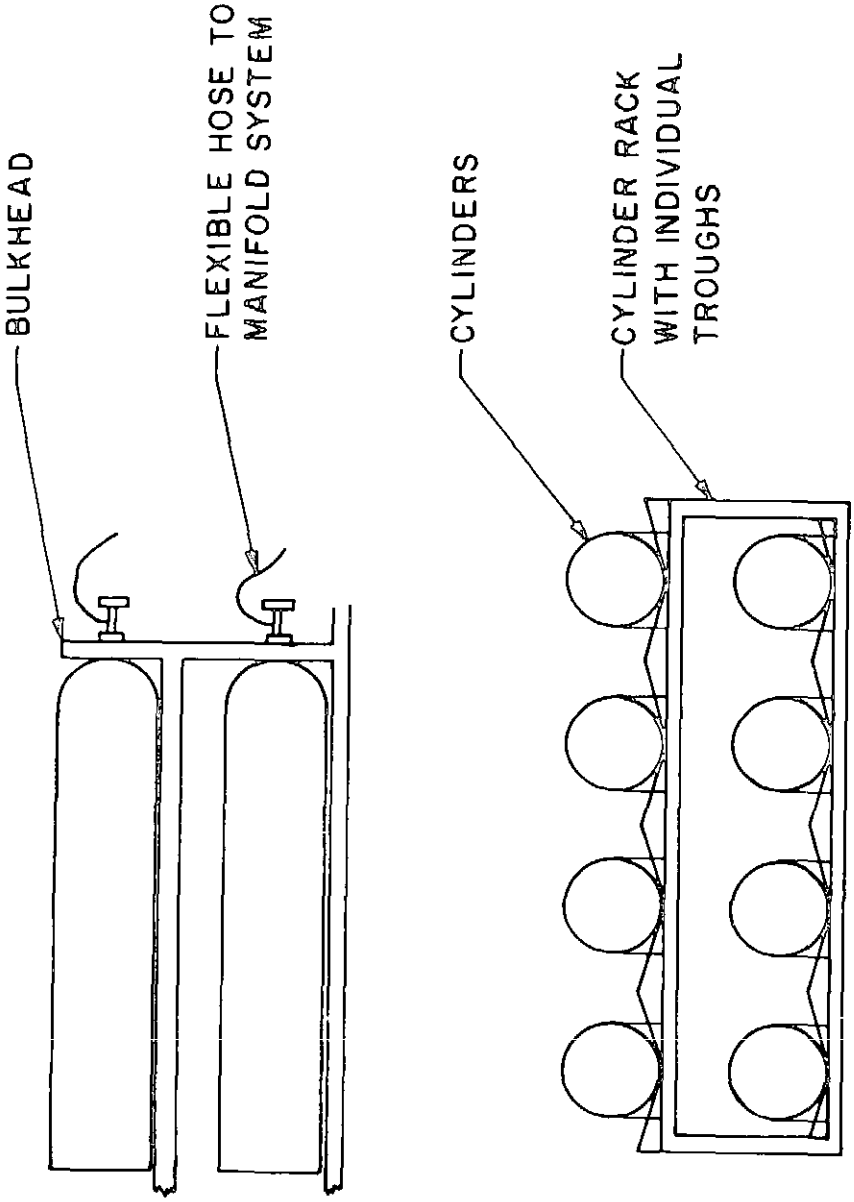


FIGURE 1. Flow diagram type I.

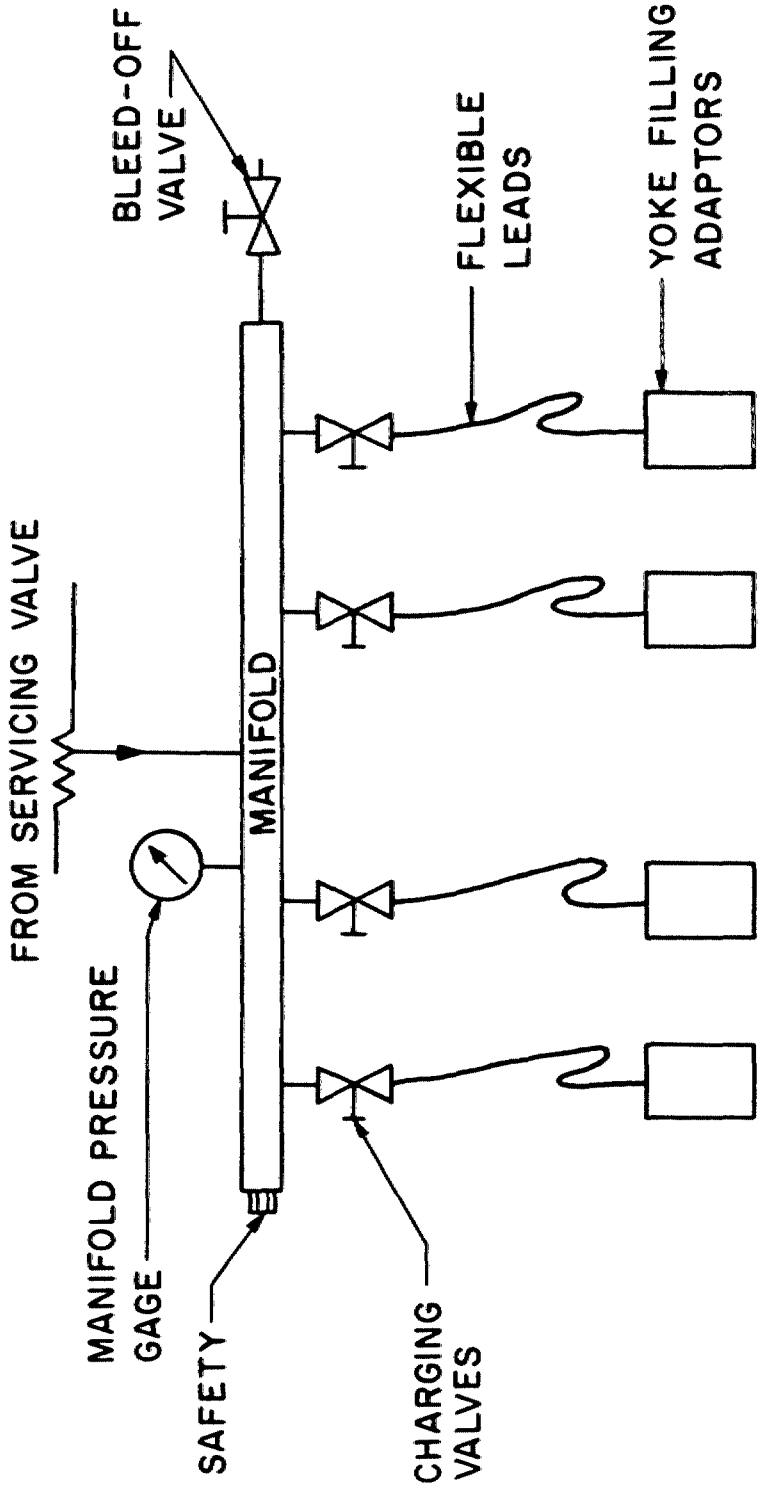
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X-3027

FIGURE 2. CYLINDER RACK TYPE I & II

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X-3028

FIGURE 3. SCUBA CHARGING MANIFOLD TYPE I

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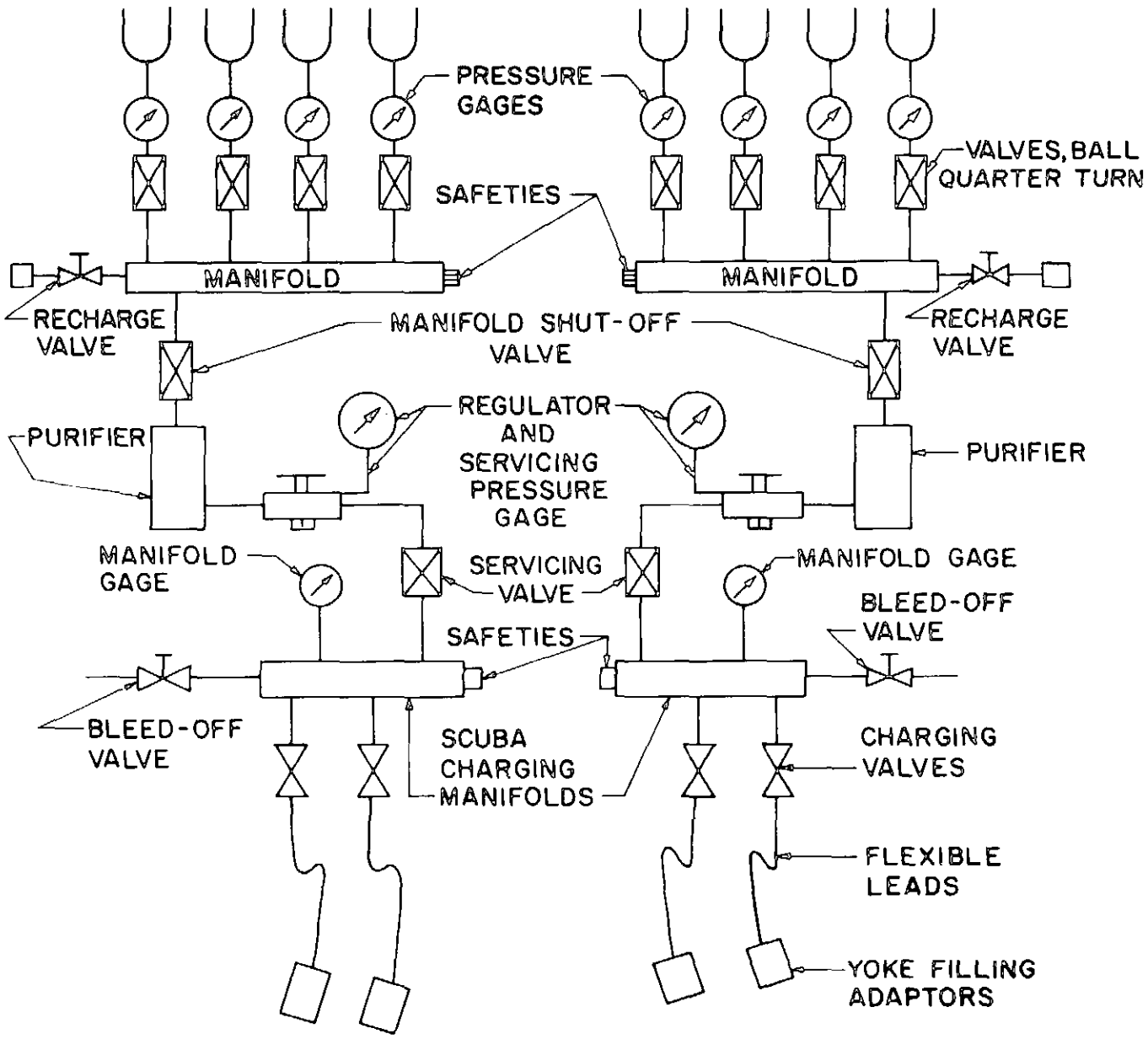


FIGURE 4. Dual air/oxygen gas flow control system type II.

X-3029A

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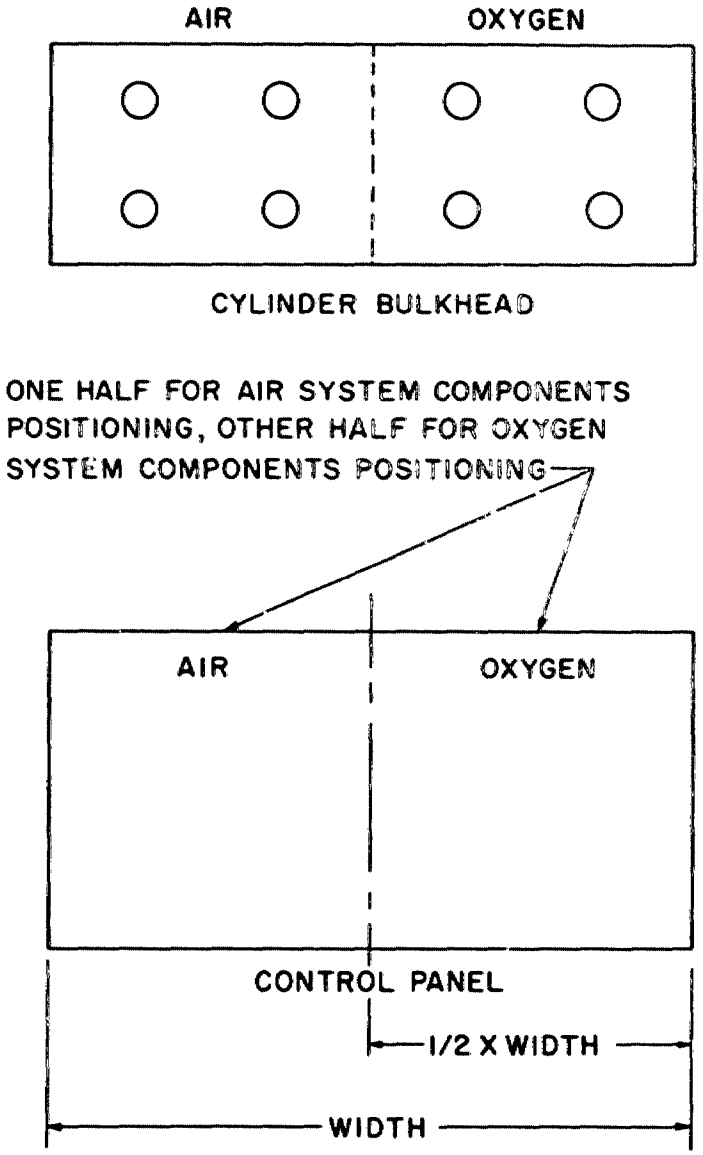
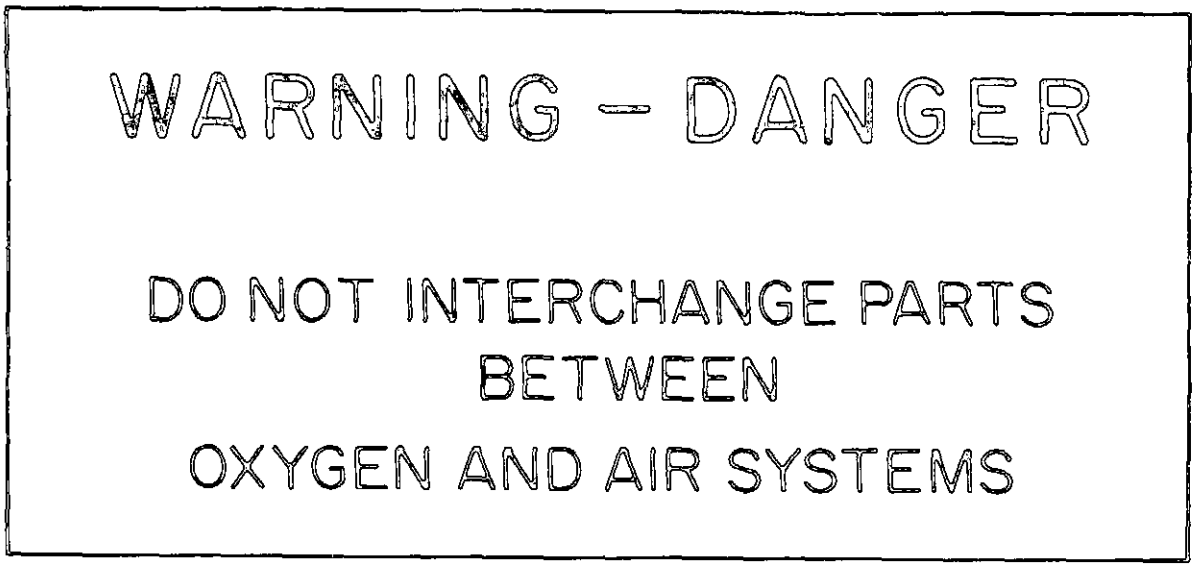


FIGURE 5. HALVING OF DUAL AIR/OXYGEN GAS FLOW CONTROL SYSTEM CONTROL PANEL & CYLINDER BULKHEAD TYPE II

X-3030

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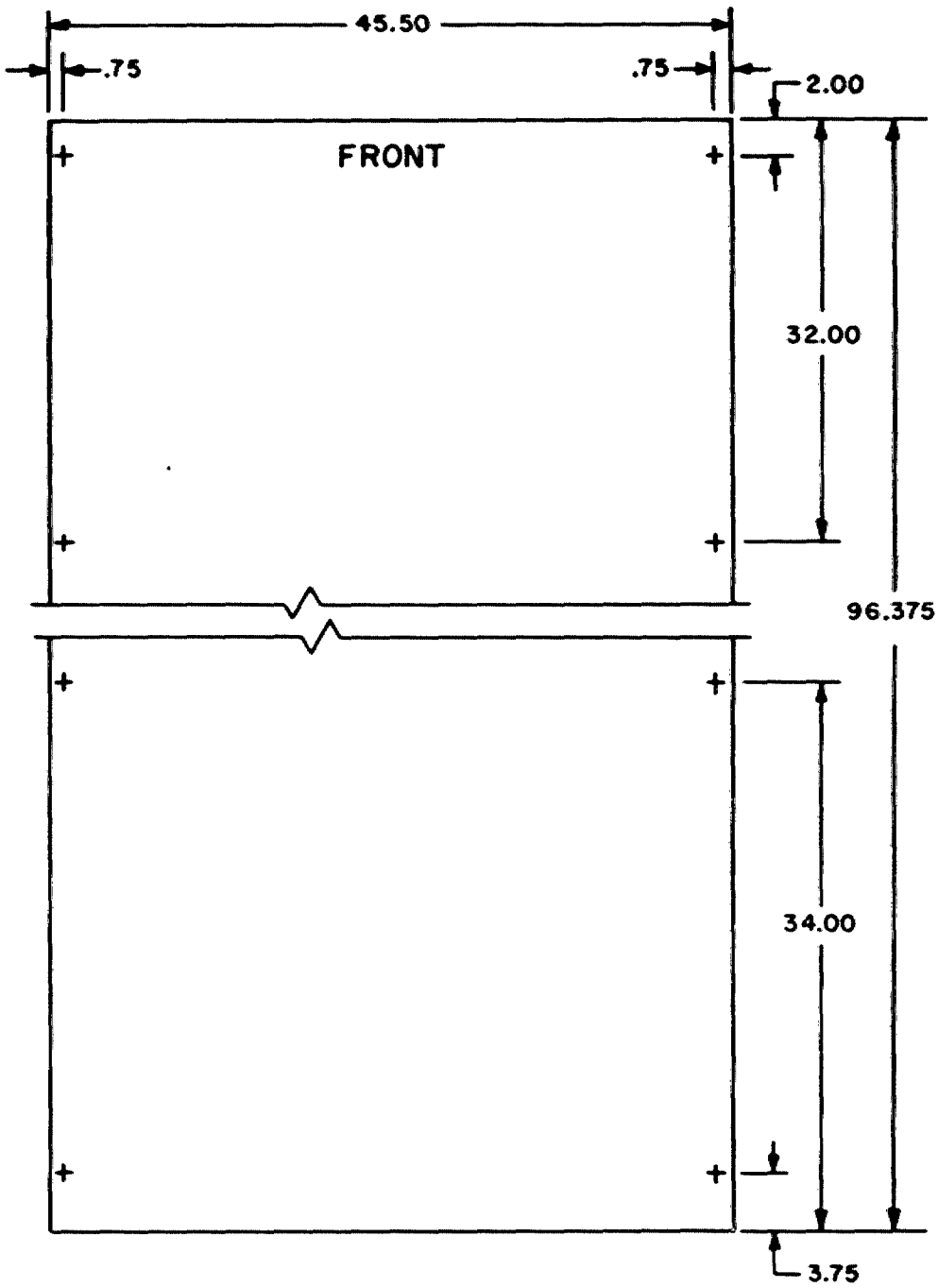


NOTE: LETTERING SHALL BE WHITE ON RED BACKGROUND

FIGURE 6. WARNING PLATE

X-3031

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NOTES:

1. + DENOTES LOCATION OF BOLT HOLE.
2. HOLE SHOULD ACCEPT A .438 DIA BOLT.
3. ALL DIMENSIONS ARE IN INCHES.

FIGURE 7. Trailer bolt hole locations.

X-4430

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-52881A(ME)	Servicing Units, Compressed Gas, Mobile

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 32 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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